

Clinic no-shows: fixing the reminder that quietly stops, before predicting anyone

Imagine a municipal health secretariat saying: “20.2% of our patients don't turn up. Build us a model that predicts no-shows so we can overbook.” That request is my own invention. The data is real: 110,527 appointments from the public health system in Vitória, Brazil, between 29 April and 8 June 2016 (the Kaggle “Medical Appointment No Shows” dataset). After cleaning: 110,516 appointments, 62,296 patients, 81 clinic neighbourhoods, and roughly **826 empty slots every working day**.

I didn't start with the model. I wanted to know where in the workflow appointments actually get lost. The short answer: the SMS reminder sometimes just stops. On 7 of the 27 days, not one patient booked in advance got a reminder, and nothing in the data suggests anyone noticed. I haven't interviewed anyone, so wherever I'm guessing I've tried to say so.

Who is involved

Who	What they want, and what limits them	What I would ask them first
Health secretariat (owns the problem, pays)	More patients seen by the same doctors. No budget for extra staff.	What would make you stop this after a month?
Front desk and schedulers	A full day and fewer idle doctors. They can't phone every patient.	Walk me through the last day half the list didn't come.
IT and the SMS vendor	Own the nightly reminder job. Every SMS costs money.	Where is the log of what was sent on 16 May?
Doctors	Lose consultation time to empty slots, and don't like overbooking.	What do you do when three patients don't show?
Patients	Long waits for a slot. Many are on welfare, work shifts, have no car.	How did you know about your appointment?

What happens now



The reminder step is where it goes wrong. It is a batch job. The only trace it leaves is a yes/no flag on the appointment: no send time, no delivery status, no phone number. So when the job fails, the record looks the same as one where the patient simply didn't get a message. The appointment's time of day, cancellations and reschedules aren't recorded at all.

What I know, and what I am assuming

Things the data shows

- Same-day bookings no-show 4.6%. Bookings 3+ days ahead no-show 29.7%.
- No booking made less than 3 days ahead ever got an SMS.
- SMS coverage for eligible bookings is 82% on a normal day, and **zero** on 9 May, 13 May and every day of 16–20 May.
- 24,536 eligible bookings got no SMS: 16,581 on those outage days, 7,955 on normal days.
- Rain doesn't seem to matter: 20.3% on dry days, 20.4% light rain, 18.8% heavy (only 2 days).

Guesses that need checking

- The request and the stakeholders above.
- That the 3-day rule is deliberate policy. I worked it out from the data; nobody told me.
- That the outage days had nothing to do with which patients booked, which is what lets me compare them.

Still unknown

- Were the SMS really not sent on those days, or just not recorded? Page 2 suggests not sent.
- Why about 18% of eligible bookings miss out even on normal days. My guess is missing phone numbers.
- Why there's no data at all for 23 May and 27 May. The holiday API only explains Corpus Christi (26 May).
- Whether the 16,200 rows where one patient has several appointments on the same day are double bookings or two specialties.

The problem, as I would put it

Patients who book 3+ days ahead in Vitória's public clinics miss 29.7% of their appointments. The SMS reminder meant to prevent this reaches only 59% of them, and it failed completely on 7 of 27 days because nothing checks it. Within eight weeks, and without changing booking rules, capacity or staffing, I would want every day's reminder coverage at 95% or more, an alert the same day it drops, and the advance-booking no-show rate under 28%.

How I would measure it

Type	Measure	Today	In 8 weeks
Main one	No-show rate for appointments booked 3+ days ahead	29.7%	under 28%
Supporting	Share of eligible bookings reminded, checked every day	59%	95%+ every day
Supporting	Days where the reminder job fails, and how fast anyone notices	7 of 27, never	none, same day
Supporting	Reminders with a send time and delivery status on record	none	all
Supporting	Weekdays missing from the data with no explanation	2	0
Guardrail	Same-day no-show rate (never reminded, so it shouldn't move)	4.6%	unchanged

Why 28% and not something lower: getting rid of the outages is worth about 1.1 points (around 654 no-shows), and closing the normal-day gap maybe another 0.6. Reminders won't fix most no-shows, but they are the part that is clearly broken, and anything smarter would depend on them.

What the data shows

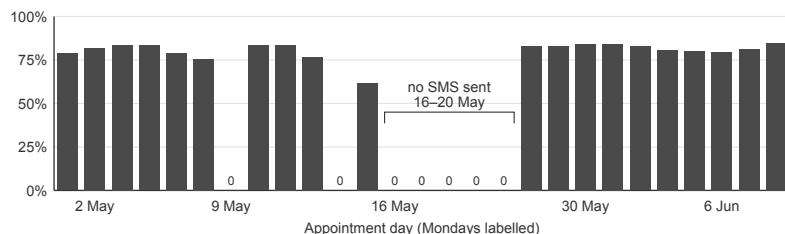


Figure 1. Share of 3+ day bookings that got an SMS, by appointment day.

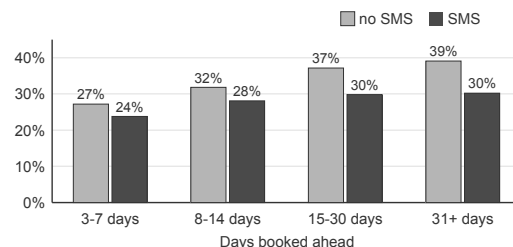
Figure 1 is what made me change my mind about the problem. Coverage sits around 80% and then drops to nothing on 9 May, 13 May and the whole week of 16 May. Since the patients booked for those days weren't chosen any differently, that is close to an experiment:

No-show rate	Normal days	Outage days	Change
Booked 3+ days ahead (should get an SMS)	28.6% (43,436)	32.5% (16,581)	+3.9 pts
Booked 0–2 days ahead (never get one)	8.93% (37,122)	8.92% (13,377)	0.0 pts
Difference, i.e. the effect of losing reminders			+3.9 pts

The patients who never get reminders didn't change at all. Only the ones who should have been reminded got worse, by about 654 extra empty slots over 7 days. Had the messages gone out and just not been logged, nothing would have changed, so I'm fairly confident they didn't go out.

There is also a trap here. Look at the whole dataset naively and patients who got an SMS no-show *more* (27.6% against 16.7%), because only long-lead bookings get reminders and long-lead bookings no-show more anyway. Figure 2 splits by lead time, and within every band the SMS group does better. A model trained on the raw data would learn that reminders make things worse.

Figure 2. No-show rate with/without SMS, normal days.



Where the data comes from

What I need	Where I got it	Where the truth lives	How far I trust it
Booking and slot date	Appointments CSV (hash-checked)	Scheduling system (IT)	Fine, but no time of day
Reminder sent?	SMS_received flag in the same CSV	SMS vendor log (don't have it)	Weak: no time or status. The key gap
Patient came?	No-show column in the CSV	Front-desk check-in	Fine, but "No" means they came
Clinic open?	Nager.Date holidays (JSON API)	Municipal calendar	National only; explains 1 of 3 missing weekdays
Weather	Open-Meteo archive (JSON API)	Just context	Modelled, not a station; fine for a test

How the pipeline works

- Retrieve:** hash-check the CSV; save each API response untouched with a timestamp. Retries on 429/5xx; if an API is down, use the last saved copy and mark the run stale.
- Validate:** four checks stop the run (columns, row count, unique IDs, outcome values). Impossible rows go to quarantine; doubtful ones are flagged and kept.
- Model:** SQLite tables for patients, appointments, reminders (the intervention), outcomes and a calendar, with a foreign-key check.
- Metrics:** 13 SQL queries. The run fails if a rate falls outside 0–1 or the groups don't add up.

Each run writes a log and a manifest (sources, hashes, row counts) and exits with a code naming the failed step. Six unit tests cover the cleaning rules.

Found and handled: 11 rows quarantined (booked after the appointment; ages –1 and 115), the backwards outcome label, misspelled columns, Handcap being a 0–4 count, and same-day bookings coming out as –1 days. The code, raw data and run logs are at github.com/abdurrahmaan11265/fde-clinic-noshow-pipeline.

What I would build first

In the first eight weeks

- A daily count of eligible bookings vs reminders sent, alerting the same day it falls under 95%.
- The vendor's send and delivery log pulled into the pipeline, replacing the flag.
- A list of eligible patients with no phone number, for the front desk.
- A look at the batch cut-off: 3-day bookings get 53% coverage, against about 85% at 4–5 days.

Deliberately not doing

- A no-show model or overbooking. The data would teach it the wrong lesson about SMS.
- Wait times, even though they matter more (4.6% against 29.7%). That needs capacity data I don't have.
- Chasing the 3,222 repeat no-show patients (5.2% of patients, 35% of no-shows) until reminders work.
- Replacing the scheduling system.

What I would want proved before going further

Question	How I'd check	Worth carrying on if
Were the SMS really not sent on those days?	Vendor logs for 9, 13, 16–20 May; ask IT	The logs show the gap
Does fixing it actually help, going forward?	Turn on monitoring in half the clinics first and compare	At least 1 point lower in 4 weeks
Why do 18% miss out on normal days?	Check the patient registry for phone numbers	One fixable cause covers most

Why this is the right problem to start with

It is the one step in the workflow I can show is broken, and the outage days show that fixing it helps. It is cheap, a count and an alert, and easy to undo if I've got it wrong. And it has to come first: a prediction model would learn from data this step has already corrupted, and whatever it decided to do (an extra message, a call) would go out through the same job that fails unnoticed. So I would fix that first, then see what is left to predict.